



KSP PART EXPLORER

User Manual

Browse • Inspect • Filter • Compare

A reference guide for players and KSP mod developers

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Tip: Word's Navigation pane can also be used to jump between the heading sections in this manual.

1. About PartExplorer

PartExplorer is a KSP1 add-on that is used to browse through, examine, filter out, and compare parts currently installed in your game. PartExplorer loads KSP's list of parts from the game itself after applying all changes that are made by ModuleManager, thus showing you actual configuration of KSP rather than just original part files.

The main Parts screen is a combination of a searchable table and an info panel. You can filter out the list based on the mod that created it, its category in KSP, text, capabilities of SCANSat and, when in the VAB or SPH, currently active filters in KSP's editor. Parts may also be compared and put directly into the editor.

Who is it for?

PartExplorer helps when making decisions among similar parts while playing the game normally, as well as while developing mods for checking of loaded values, internal IDs, ModuleManager outcomes, resource capabilities, science setup, or the path utilized by KSP.

Core capabilities

- Browse parts from one mod or all installed part mods.
- Filter by KSP category and free-form text.
- Use SCANSat-specific filters when scanner parts are present.
- Sort the list and add optional data columns.
- Inspect stock module, resource, science-module, and SCANSat data.
- Select multiple parts and compare their values side by side.
- Highlight different values, hide identical rows, mark low/high values, and show deltas from the first selected part.
- Review installed mods and detected science support.
- Add a selected part directly to the VAB or SPH.
- Reuse cached part data when reopening PartExplorer to avoid reparsing the entire part database.

2. Installation and Requirements

KSP compatibility

PartExplorer 0.1.69 is configured for Kerbal Space Program 1.12.x.

Required dependencies

Dependency	Purpose
ToolbarController	Provides stock/Blizzy toolbar integration and placement.
ClickThroughBlocker	Prevents mouse input on the PartExplorer window from passing through to the KSP scene.

3. Opening and Using the Window

Toolbar button



Click the PartExplorer toolbar button to show or hide the window. ToolbarController determines whether the button appears on the stock Application Launcher, the Blizzy toolbar, or another supported placement.

Window behavior

- The window can be dragged and resized.
- On the Parts page, the vertical divider between the list and Details pane can be dragged to change the pane widths.
- The window automatically hides during scene changes.
- The window hides while KSP is paused and is restored when the game is unpaused if it was open before pausing.
- Window size, position, pane width, filters, column choices, comparison options, and skin choice are remembered between sessions.

Main tabs

Tab	Purpose
Parts	Browse, filter, select, and inspect loaded parts.
Compare (n)	Compare the n parts currently selected for comparison.
Mods	Review loaded mods, part counts, and science support.
Settings	Control appearance, visible fields/panels, and optional table columns.

Parts-page layout

LEFT: PARTS LIST	RIGHT: DETAILS
Comparison checkboxes, thumbnail, part name, mass, active sort value, and any additional columns.	Description, preview, part information, stock module data, resources, science data, and SCANSat data for the selected row.

4. Parts Tab

Mod and category filters

The Mod selector is built from the first folder in each loaded part's KSP partUrl. In most installations this is the top-level folder under GameData, such as Squad, SCANSat, or the folder name used by another part mod. Choose All installed mods to browse everything together.

The Category selector contains Any plus the KSP categories that actually occur in the currently selected mod. Parts whose KSP category is none are ignored. Changing the mod refreshes the available category choices.

Text filter

The Filter field searches across the part data that PartExplorer has loaded. It can match part names and other displayed or searchable metadata, including:

- Part title and internal ID
- Mod name and KSP category
- Loaded part path
- Stock module and resource details
- Configured science-module fields and resources
- SCANSat scan information when present

The Clear button clears the text filter and resets the SCANSat-specific filters to Any. Mod and Category selections are not cleared by that button.

Filtered only

Filtered only works best for VAB or SPH. In case you enable the feature in either of these two places, the PartExplorer uses KSP's active editor filters together with its own filters of Mod, Category, Text, and SCANSat. This feature has an impact on the parts that appear in the Comparison page.

SCANSat filters

Only when the selection of Mod is SCANSat will the SCANSat filter controls become visible. This is the case only when the currently filtered list has SCANSat scanner parts. The selected SCANSat filters will be available the next time SCANSat becomes pertinent, but the hidden SCANSat filters will not be applied elsewhere.

Filter	Choices
Biome	Any / Yes / No
Altimeter	Any / Low / High
Visual	Any / Low / High
Resource	Any / Low / High
Anomaly	Any / Yes / No
Daylight	Any / Yes / No

Sorting

Sorting is done using the Sort Control Buttons labeled Part, Cost, Mass, EC/s, and Science. Clicking on the selected sorting button reverses the sort from ascending to descending order. Table headers that appear can also be used to sort the table.

Active sort value

Name and Mass are always present in the parts table. When you sort by another value that is not already enabled as an additional column, PartExplorer temporarily shows that sort value as an extra column so the ordering remains easy to understand.

Parts list

The column headings remain pinned while the part rows scroll vertically. Horizontal scrolling keeps the headings synchronized with the data rows.

Area	Action
Compare checkbox	Select or deselect this part for the Compare tab.
Thumbnail	Shows the part image. Hover to open a larger rotating 3D preview when live part geometry is available.
Part name	Selects the row and updates the Details pane.
Mass	Shows the loaded dry mass in tonnes.
Other columns	Shows the active sort value and any columns enabled in Settings.

Comparison selection controls

Button	Effect
Select visible	Adds every currently visible part to the comparison selection.
Clear visible	Removes currently visible selected parts, without affecting selected parts hidden by filters.
Clear	Clears the entire comparison selection.

The status line below the list shows the number of parts currently visible and the total number selected for comparison.

Right-click menu

Right-click any part row for quick actions:

Command	What it does
Add to editor	Adds the part in the VAB/SPH. Disabled outside the editor or if the part cannot be resolved in the loaded editor database.
Add to Compare / Remove from Compare	Changes comparison selection without using the leftmost checkbox.
Copy part ID	Copies the internal KSP part ID to the system clipboard.
Copy path	Copies the loaded KSP part path to the system clipboard.

5. Details Pane

Selecting a part name in the list updates the Details pane. The available panels depend on the modules and data present on that part and on the visibility choices in Settings.

Description and part image

The Description section shows the loaded part description. The image area shows the available thumbnail; hovering over the image can display a larger rotating 3D preview generated from the loaded KSP part.

Part Information

The Part Information panel can show the internal ID, mod, category, cost, mass, maximum temperature, impact tolerance, and G tolerance. The loaded part path can be shown separately by enabling Show part path in Details in Settings.

Stock information panels

Panel	Examples of information shown
Cargo Part Info	Packed volume, stackability, and related cargo values.
Command	Minimum crew, hibernation settings, and command-module resource use.
Data Transmitter	Antenna type/power, packet settings, resource use, range, and combinability.
Probe Control Point	Minimum crew and multi-hop capability.
Reaction Wheel	Pitch, yaw, and roll torque and resource consumption.
SAS	SAS service level.
Resources	Stored resources with loaded/default amount and maximum capacity.

Science Modules

When PartExplorer recognizes a configured science PartModule, the Science Modules panel shows the fields defined for that module and any detected input/output resources. ScienceMods support is extensible; see Section 11.

SCANsat panels

Panel	Information
SCANsat	Electric Charge use, field of view, daylight requirement, and science value.
Scan Types	Biome, altimetry, visual, resource, and anomaly scan capabilities.
Altitude Range	Minimum, optimal, and maximum scanner altitudes.

Add Part to Editor

If PartExplorer is launched while being in the VAB or SPH and the chosen part that is loaded can be located within the editor database of KSP, the user can use the function Add Part to Editor that appears in the Details tab. PartExplorer uses the standard editor spawning procedure of KSP.

6. Comparing Parts

Use the checkbox at the left of a Parts row, the row's right-click menu, or Select visible to build a comparison set. The Compare tab caption shows the current number of selected parts.

Compare table

Every chosen part appears in its own column. In the leftmost column called Detail, the property name appears and the values of the selected parts appear in each row. If there are no values, an em dash appears. In the row labeled Description, wrapping of text is allowed along with additional vertical spacing.

In the VAB or SPH, the part-name buttons located at the top of the comparison columns enable the addition of the respective part to the editor. The red X deletes the respective part from the comparison list.

Comparison options

Option	Behavior
Highlight differences	Highlights cells on rows whose displayed values are not identical across all compared parts. The current highlight color is RGB 239, 203, 5.
Differences only	Hides rows whose displayed values are identical for every selected part. Available when comparing more than one visible part.
Mark low/high	For simple numeric rows with matching units, marks the lowest and highest values.
Deltas vs first	For simple numeric rows with matching units, appends each value's difference from the first selected part.

Column width

Mode	Best use
Compact	Fits more part columns on screen; best for mostly short values.
Normal	Balanced default width.
Wide	More room for descriptions and long module/resource values.

The chosen column-width mode is saved between sessions.

Filtered comparison

When Filtered only is enabled, Compare shows only selected parts that also pass the current PartExplorer filters and KSP editor filters. Hidden selections are not deleted. Turning Filtered only off restores them to the comparison table.

7. Mods Tab

The Mods tab summarizes the mod folders represented in the loaded part database. It shows the number of parts attributed to each mod and whether PartExplorer detected configured science-module data for parts from that mod.

Click a mod name in this table to apply that mod as the Parts-page Mod filter and return directly to the Parts tab.

Column	Meaning
Mod	Top-level mod folder name derived from loaded part URLs.
Parts	Number of loaded parts attributed to that mod.
ScienceMods support	Detected science support and the number of affected parts, or None detected.

8. Settings

Interface

Setting	Effect
Use alternate KSP skin	Switches PartExplorer between its standard and alternate KSP-style skin.
Show part path in Details	Displays the loaded KSP part path in the Details pane.

Details pane fields

These switches control individual general fields in the Part Information panel: ID, Cost, Mass, Max temperature, Impact tolerance, and G tolerance.

Details panels and Compare rows

The same panel categories can be enabled independently for Details and Compare. Hiding a panel in Details does not automatically remove it from Compare, and vice versa.

Available category	Typical content
Part Information	General part values.
Cargo Part Info	Cargo inventory metadata.
Command	Command module behavior.
Data Transmitter	Antenna/transmitter data.
Probe Control Point	Probe-control capability.
Reaction Wheel	Torque and resource use.
SAS	SAS level.
Resources	Stored resources and capacity.
Science Modules	Configured science fields/resources.
SCANSat	Scanner EC, FOV, daylight, science.
Scan Types	Biome/altimetry/visual/resource/anomaly.
Altitude Range	Minimum/optimal/maximum scanner altitude.

Additional data columns

Optional columns can be added to the Parts table. Click any visible heading to sort by that column; click the same heading again to reverse the direction. Clear Additional Columns removes all optional columns.

Column	Value
ID	Internal KSP part ID
Cost	Part cost
EC/s	SCANSat Electric Charge use
Science	SCANSat science value

Column	Value
FOV	SCANSat field of view
Daylight	Whether daylight is required
Min Alt	Minimum scan altitude
Opt Alt	Optimal scan altitude
Max Alt	Maximum scan altitude
Max Temp	Maximum temperature
Impact	Impact tolerance
G Tol	G tolerance
Biome	Biome scan capability
Altimetry	Altimetry capability
Visual	Visual scan capability
Resource	Resource scan capability
Anomaly	Anomaly scan capability

9. Editor Integration

PartExplorer detects whether KSP is currently in the VAB or SPH. Editor-only commands are enabled only when an AvailablePart can be resolved from the loaded part database and EditorLogic is available.

Ways to add a part

- Select a part and click Add Part to Editor in Details.
- Right-click a part row and choose Add to editor.
- On the Compare page, click the part name at the top of a comparison column.

PartExplorer invokes KSP's normal SpawnPart behavior. After adding from Details or the context menu, the reference window closes so you can place the new part.

Using KSP editor filters

Check Enable Filtered only to ensure that the PartExplorer takes into consideration the existing filters for VAB/SPH parts. It is particularly useful if you wish to use PartExplorer for its enhanced functionality but at the same time have the results conforming to some existing filters within the editor.

10. SCANSat Support

If there is a SCANSat scanning module in the part, PartExplorer reads the live scanner settings once KSP and ModuleManager have loaded the part. It will then be possible to see the changes that ModuleManager has made to the scanner.

The supported SCANSat details include SCANSat scan modes, altitudes, field of view, sunlight requirement, Electric Charge usage, and scientific SCANSat data if available. In case live loaded part data is not available, PartExplorer can always default back to embedded SCANSat data.

Context-sensitive interface

SCANSat-only filters and panels do not clutter ordinary part browsing. They appear only when relevant scanner data is present or when SCANSat itself is selected.

11. ScienceMods Support

PartExplorer can display fields from additional science PartModules without recompiling the plugin. Definitions are loaded from .cfg files in:

GameData/PartExplorer/PluginData/ScienceMods

Definition format

Each definition names a PartModule and one or more fields. Field numbers determine display order. A fixed label can be supplied with `_Title`, or another field can supply the label with `_TitleField`.

```
{
  ModuleName = MyScienceModule

  expField_1 = experimentID
  expField_1_Title = Experiment ID

  expField_2 = experimentValue
  expField_2_TitleField = experimentTitle
}
```

Resource discovery

If two modules are matching science modules, then PartExplorer will also verify common resource declaration statements like `INPUT_RESOURCE`, `OUTPUT_RESOURCE`, `INPUTRESOURCE`, `OUTPUTRESOURCE`, `RESOURCE`, and supported field based resource declarations. The resource name along with its rate, ratio, or amount is displayed, wherever applicable.

Reloading definitions

ScienceMods' definitions will be updated when PartExplorer updates its part database. An ordinary cache refresh operation does not update all parts. When there is any change to the count of the loaded-parts database, then PartExplorer will update the database and definitions..

12. Performance, Caching, and Persistence

Part-data cache

The first time that PartExplorer requires part information while working within KSP, it constructs its own internal databases using the part information contained in the KSP database, constructs lookup tables for use with editor integration, and loads the ScienceMods information. This process may be slow on heavily-modded installations.

After that successful build, when PartExplorer is closed and then reopened, the data in memory is used rather than parsing each installed part all over again. Before reuse, PartExplorer runs a quick validation test. The cache is rebuilt automatically if KSP replaces the LoadedPartsList or the number of parts changes.

Expected behavior

A first open after loading of KSP may consume more time than subsequent opens. Subsequent opens within the same KSP session must take less time since the recorded data from the previous loaded part is already cached.

Editor-filter cache

The caching mechanism used for filter results by the KSP editor is different from the caching of part data. It is invalidated only when there is a change in "Filtered" state or "PartExplorer" must be updated with current editor filter status.

Saved preferences

PartExplorer stores its users' choices using the KSP PluginConfiguration framework. The stored information includes window size and position, pane sizes, mod/category selection, filters, extra columns, visibility of Details/Compare panel, comparison setting, compare column size, and skin setting..

13. Troubleshooting

Symptom	What to check
Toolbar button is missing	Verify that PartExplorer, ToolbarController, and ClickThroughBlocker are installed correctly. Check the ToolbarController placement/settings used by the current KSP install.
No parts are shown	Set Mod to All installed mods, Category to Any, clear the text filter, and reset SCANSat filters. In the editor, also try disabling Filtered only.
A selected comparison part disappeared	If Filtered only is enabled, the part may simply be hidden by current PartExplorer or editor filters. The comparison selection is retained.
Add to editor is disabled	The command works only in the VAB/SPH and requires the selected record to resolve to a loaded AvailablePart.
SCANSat filters are not visible	They appear only when SCANSat is selected or the current list contains SCANSat scanner parts.
Science fields are missing	Check that the module name and field names in the ScienceMods definition match the loaded MODULE configuration. A full database rebuild reloads the definitions.
First open is slow	A large modded installation can take time for the initial part-data build. Subsequent opens in the same KSP session should use the cache.
Values differ from a source part.cfg	PartExplorer reads KSP's loaded configuration. ModuleManager patches and other loading-time changes may make the final value different from the original source file.

Log information

Messages are logged using regular Unity/KSP logging functionality. In case the window cannot build information about a part, load a texture, or save settings, you should look for messages starting with [PartExplorer] in KSP.log and the Player.log.

14. Quick Reference

Goal	Fastest method
Find a known part	Set the Mod if known, then type part of its title or ID in Filter.
See only one category	Choose the Mod, then select the KSP Category.
Match the current VAB/SPH filter	Enable Filtered only.
Compare a family of parts	Filter the list, click Select visible, then open Compare.
See only differences	On Compare, enable Differences only and optionally Highlight differences.
Check which value is smallest/largest	Enable Mark low/high on Compare.
Compare change relative to a baseline	Put the baseline part first, then enable Deltas vs first.
Copy the internal name	Right-click the row and choose Copy part ID.
Copy the KSP part path	Right-click the row and choose Copy path.
Add a part without closing the comparison	In the VAB/SPH, click its part-name button at the top of the Compare column.
Inspect science support by mod	Open Mods and review the ScienceMods support column.
Show more list data	Open Settings and enable Additional data columns.

Appendix A. Bundled Science Support

PartExplorer 0.1.69 includes explicit ScienceMods definitions for the following modules/mod families:

- SCANSat — SCANexperiment
- Wild Blue Industries WBIScience 1.5.1
- Nehemiah Engineering Orbital Science 0.10.0
- DMagic Orbital Science 1.4.3
- Station Science Continued 2.6.0
- LTech Continued 1-0.5.3.2
- James Webb for Kerbal 1.12.x
- Impact 1.9.2
- Exolnstruments 0.5.1
- Tarsier Space Technology Continued 7.13
- Microbiology Expansion 0.2.0

Reusable definitions are also included for the stock KSP ModuleScienceExperiment, ModuleScienceLab, and ModuleScienceConverter modules. Mods that use these stock modules can therefore be recognized without a separate mod-specific definition.

Examples noted in the PartExplorer documentation include KrakenScience 1.0, Mkerb Inc. Science Instruments 1.1, KDEX 2.0.2, Interkosmos 0.5, and Planetside Exploration Technologies 1.0.2.